

LOSO Experiment C_right_Istm

LSTM only · Full sensors, RIGHT hand only, no Nothing

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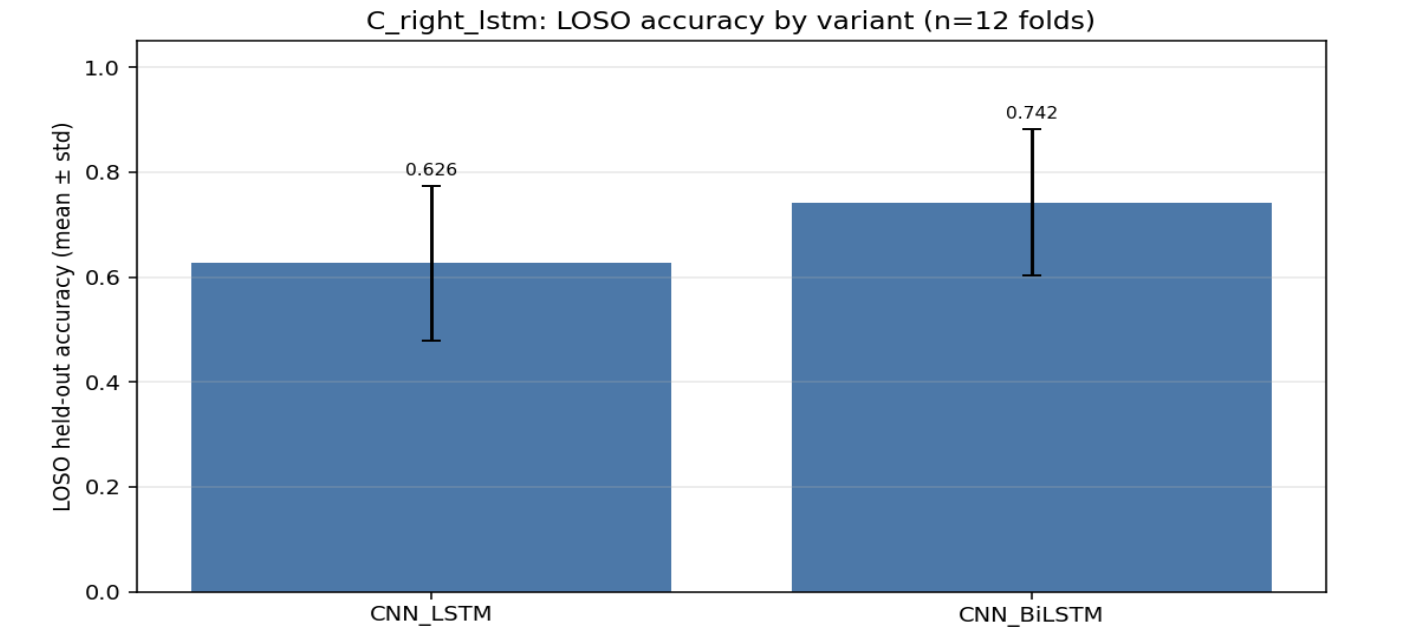
Experiment configuration

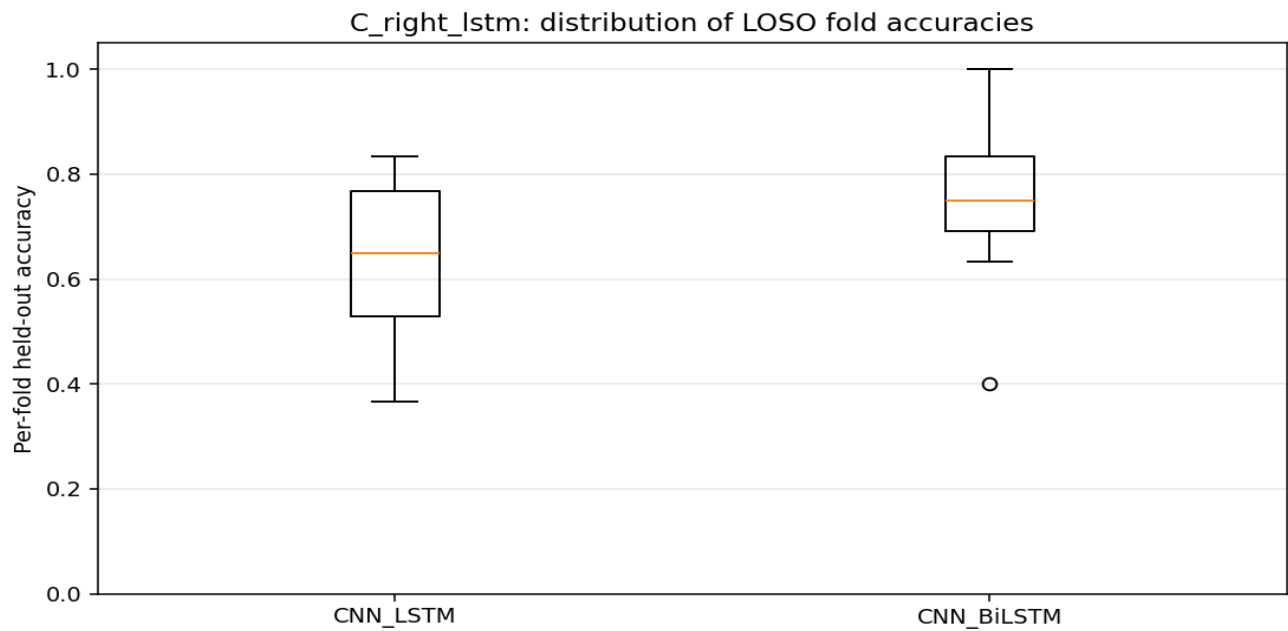
Class set	6 classes excluding Double_Nothing
Classes	Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	flex, YPR, accel
Hands	right
Sensor channels	88
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	12: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

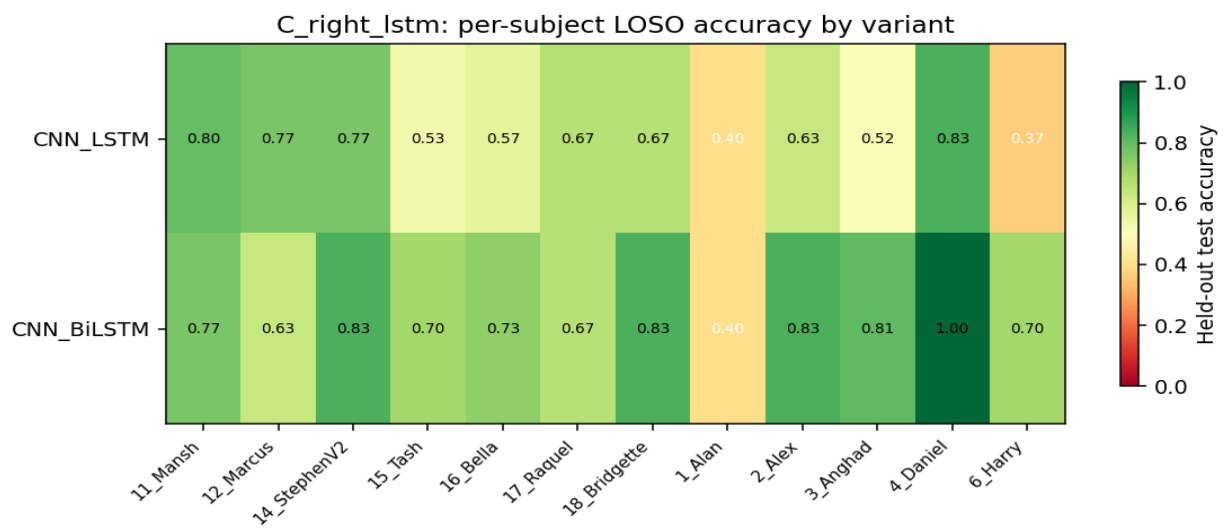
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
CNN_LSTM	57,126	0.6263	0.1468	0.3667	0.8333	0.7801
CNN_BiLSTM	94,246	0.7422	0.1398	0.4000	1.0000	0.8338

Best variant by LOSO mean: **CNN_BiLSTM**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	CNN_LSTM	CNN_BiLSTM
11_Mansh	331	0.800	0.767
12_Marcus	331	0.767	0.633
14_StephenV2	331	0.767	0.833
15_Tash	331	0.533	0.700
16_Bella	331	0.567	0.733
17_Raquel	331	0.667	0.667
18_Bridgette	331	0.667	0.833
1_Alan	331	0.400	0.400
2_Alex	331	0.633	0.833
3_Anghad	330	0.516	0.806
4_Daniel	331	0.833	1.000
6_Harry	331	0.367	0.700

Variant: CNN_LSTM

Params: 57,126

LOSO mean accuracy: 0.6263 ± 0.1468 (min 0.3667, max 0.8333)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	3/5	5/5	5/5	1/5	5/5	5/5	0.800
12_Marcus	5/5	5/5	5/5	0/5	3/5	5/5	0.767
14_StephenV2	0/5	5/5	5/5	5/5	3/5	5/5	0.767
15_Tash	0/5	5/5	0/5	5/5	5/5	1/5	0.533
16_Bella	3/5	5/5	4/5	0/5	5/5	0/5	0.567
17_Raquel	5/5	5/5	5/5	0/5	5/5	0/5	0.667
18_Bridgette	5/5	5/5	5/5	0/5	5/5	0/5	0.667
1_Alan	5/5	1/5	5/5	0/5	0/5	1/5	0.400
2_Alex	3/5	4/5	5/5	0/5	2/5	5/5	0.633
3_Anghad	0/5	5/5	5/5	1/5	0/6	5/5	0.516
4_Daniel	5/5	5/5	0/5	5/5	5/5	5/5	0.833
6_Harry	5/5	2/5	0/5	0/5	4/5	0/5	0.367
MEAN	0.650	0.867	0.733	0.283	0.700	0.533	0.626

Variant: CNN_BiLSTM

Params: 94,246

LOSO mean accuracy: 0.7422 ± 0.1398 (min 0.4000, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	5/5	5/5	3/5	5/5	0/5	0.767
12_Marcus	0/5	4/5	5/5	5/5	5/5	0/5	0.633
14_StephenV2	0/5	5/5	5/5	5/5	5/5	5/5	0.833
15_Tash	5/5	5/5	2/5	5/5	4/5	0/5	0.700
16_Bella	5/5	5/5	0/5	2/5	5/5	5/5	0.733
17_Raquel	5/5	5/5	5/5	0/5	5/5	0/5	0.667
18_Bridgette	5/5	5/5	5/5	5/5	5/5	0/5	0.833
1_Alan	4/5	2/5	5/5	0/5	0/5	1/5	0.400
2_Alex	5/5	4/5	5/5	5/5	1/5	5/5	0.833
3_Anghad	0/5	5/5	5/5	5/5	6/6	4/5	0.806
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	0/5	4/5	4/5	3/5	0.700
MEAN	0.733	0.917	0.783	0.733	0.817	0.467	0.742